Chemistry Labs

Alveary

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Science

Chemistry Labs

GRADE(S): 9-12+

About the Course

Labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method. Labs for this course are integrated into the lessons to facilitate adequate time for more involved activities and to better coordinate with the lessons.

This topic is included in the following course(s): Science: Chemistry

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12+	Chemistry Labs 1 time/week 60 min	Alveary Chemistry Lab Book

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Chemistry				
Chemistry Lessons	Chemistry Lessons Nature Walks & Scouting: Grades 1-8	Chemistry Lessons	Chemistry Lessons Nature Notebook: Grades 9-12	Chemistry Lessons Chemistry Labs

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details]

Chemistry Labs

Alveary Chemistry Lab Book

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

- [View Basic Supplies]
- [View Supply List Details]

Chemistry Labs

 Safety Goggles

Quick Links

(No Quick Links Assigned)

Click THIS text or scan the QR code for links.

Chemistry Labs

How To Approach

Introduce

• Before beginning, be sure that the basic rules of lab safety (as listed in the lab manual) are understood and obeyed.

• Every science lab has the same flow, which follows the scientific method and is guided by the lab book.

• Learners begin with an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is how hypotheses are formed.

• If learners are unsure what to do with the Introduction, it can help to dialogue with someone as they work through the ideas. Check the Chemistry Companion in the Quick Links if a video tutorial might be helpful. This year is a pilot for this video series, so videos will be added to the Quick Link throughout the year.

• Once they have had a chance to think, learners compose a prelab narration to put these introductory ideas and hypotheses into their lab notebooks. These prelab narrations may seem short and even incomplete if learners are new to keeping a lab notebook, but that is okay.

• If learners have difficulty or are easily frustrated, then consider allowing a digital notebook for typing, a scribe, or use of assistive technology, as appropriate.

• After they complete their prelab narration, learners should collect the listed materials, being sure to let teachers know if something is missing or to remind the teacher if something needs to be purchased at the store.

Lab Procedure

• Learners complete the procedure as instructed in the lab book.

• The lab gives instructions for using the lab notebook to create tables and diagrams.

Analysis & Conclusions

• The last step in the lab is to analyze the data and observations and draw conclusions from them. This postlab narration is prompted in the lab manual. Similar to the prelab narration, the concluding or postlab narration is a chance to think about and put into words. This time, learners are considering what they learned from the lab and what they could learn more about, if they were to continue.

• Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.

• Depending on the interest of the learner and the priorities of the teacher, the learner might be encouraged to spend more time on those ideas of what more they could learn or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we

	could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.